



Automated Computing Machinery SL - Instrumentation Division

GPTR

3Phase Grid Generator and 4 quadrants Load Simulator

Product Description and Specifications



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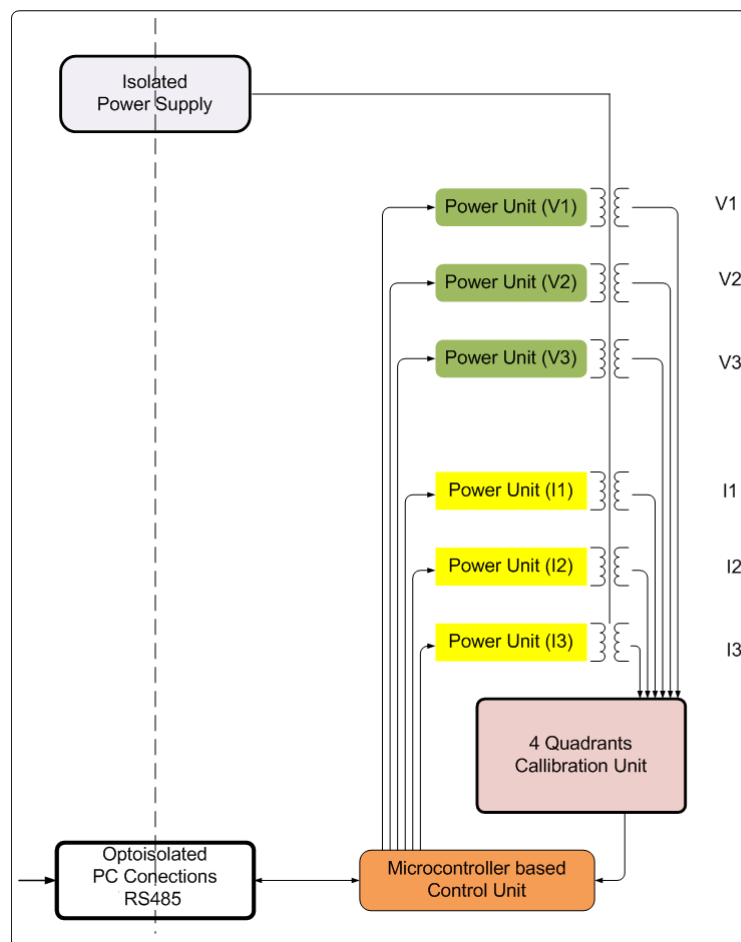
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Description

This equipment is a subsystem on its own, aimed to serve as test boiler plate for control systems that governs the behavior of equipment that must work connected to a 3Phase grid. I.e. wind generators, cogeneration installations, etc.

To achieve that it includes two sets of three generators, synchronous among them: three for the actual grid generation (V1, V2 and V3) and three to emulate the currents measured at a given load through current transformers (I1, I2, I3). See the block diagram below. The parameters of each generator (voltage or current, relative phase to V1 generator) as well as the common frequency are controlled by an external program through simple commands.



The microcontroller generates the control signals for each one of the generators, sharing a common frequency generator. Each generator has an independent output transformer.

The equipment includes a certified power measuring unit (3phase, four quadrants, class 0.5%) that is used for on-line calibration of each generator.



Characteristics:

Voltage generators: 0 to 420 V rms, in 1 volt steps ⁽¹⁾⁽³⁾

(* built-in over-current tripping behavior adjusted to 12 mA rms in each channel, for safety reasons)

Current generators: Two options:

- a) 0 to 1 A. rms with 4 ohms output impedance, in steps of 2 mA rms. ⁽³⁾
- b) 0 to 5 A rms with 0,4 ohms output impedance, in steps of 10 mA rms. ⁽³⁾

Frequency: adjustable between 39 to 65 Hz, in 1 Hz steps. ⁽²⁾

Phase adjustment: The phase of each generator can be specified relative to the V1 channel, between 0 to 360°, in 1 ° step. ⁽⁴⁾

Calibration: the equipment includes an independent Grid Analyzer that is used to measure the real generated signals. Its measurements are read by the microcontroller and used to calibrate internally the parameters for each generator. This is done with a specific command. This unit is a four quadrants, class 0.5 %, provided with certified calibration or similar device. There is

- (1) This is the warranted margin. Typically the upper limit is in the range of 450 V rms.
- (2) Output transformer are design for the 45 to 55 Hz range, so when frequency is changed out of that range, it is recommended to issue the self calibration command. Typically, though, frequency response is flat from 39 to 61 Hz.
- (3) See discussion on slew rate in the “Commands” section
- (4) See discussion on changes for phase values in the “Commands” section



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Commands

The GPTR is controlled via a RS485 line, half-duplex, at 9600 baud rate. It accepts a number of commands that control its behavior and the values of the generated signals. These are lowest level commands.

The following list correspond to the release “GPTR_01 REV_02 (27/11/2005) VSW0.05 Transient generation” and “GPTR_01 REV_01 (02/10/2005)” of the firmware used by the microcontroller.

Note: all data values are expressed with 4 digits left padded with “0”. I.e. “10 V” is specified as “0010” in the corresponding command.

1. Command for controlling the RS485 interface:

Command description: Set Configuration

Purpose: To change the RS485 address to which the unit will respond to.

Format: “%(OldAddr)(NewAddr)(Checksum)<CR>”

Where: “%” is ASCII character %, “<CR>” is ASCII character for Carrier return (13 hexa)

(OldAddr) : two hexadecimal digits corresponding to the actual RS485 address of the unit.

(NewAddr) : two hexadecimal digits corresponding to the desired (new) RS485 address of the unit.

(Checksum): 2 hexadecimal digits corresponding to the checksum of the previous “%(OldAddr)(NewAddr)” digits.

Possible answers:

If everything OK (ACK) “>(OldAddr)(Checksum)<CR>”

If some error (NACK) “?(OldAddr)(Checksum)<CR>”

2. Command for setting the values of the signals:

General format: “\$(Addr)W(Variable)(Value)(Checksum)<CR>”

Where :

(Addr) is a two hexadecimal digits corresponding to the unit address in the RS485 line

(Variable) is a three letters indicator of the signal to be set, according to the following list

Signal	Signal description
--------	--------------------



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VE1	Voltage for Phase 1
VE2	Voltage for Phase 2
VE3	Voltage for Phase 3
IE1	Current for current transformer associated to Grid Phase 1
IE2	Current for current transformer associated to Grid Phase 2
IE3	Current for current transformer associated to Grid Phase 3
FRQ	Frecuency
FV2	Phase angle of V2 respect to V1
FV3	Phase angle of V3 respect to V1
FI1	Phase angle of I1 respect to V1
FI2	Phase angle of I2 respect to V1
FI3	Phase angle of I3 respect to V1
RTI	Relation of the current measurement transformers used in the actual installation ⁽¹⁾
CV1	Calibration value associated to V1 generator
CV2	Calibration value associated to V2 generator
CV3	Calibration value associated to V3 generator
CI1	Calibration value associated to I1 generator
CI2	Calibration value associated to I2 generator
CI3	Calibration value associated to I3 generator
TTR	Option HT : Total transient duration, in cycles (10...3000) ⁽²⁾
RS1	Option HT : Rising ramp for V1, in Volts/cycle (1..1000) ⁽³⁾
RS2	Option HT : Rising ramp for V2, in Volts/cycle (1..1000)
RS3	Option HT : Rising ramp for V3, in Volts/cycle (1..1000)



RS4	Option HT : Rising ramp for I1, in Amps/cycle (1..1000)
RS5	Option HT : Rising ramp for I2, in Amps /cycle (1..1000)
RS6	Option HT : Rising ramp for I3, in Amps /cycle (1..1000)
RB1	Option HT : Falling ramp for V1, in Volts/cycle (1..1000)
RB2	Option HT : Falling ramp for V2, in Volts/cycle (1..1000)
RB3	Option HT : Falling ramp for V3, in Volts/cycle (1..1000)
RB4	Option HT : Falling ramp for I1, in Amps/cycle (1..1000)
RB5	Option HT : Falling ramp for I2, in Amps/cycle (1..1000)
RB6	Option HT : Falling ramp for I3, in Amps/cycle (1..1000)
CT1	Option HT : Multiplication factor applied to V1 in % (0..100) ⁽⁴⁾
CT2	Option HT : Multiplication factor applied to V2 in % (0..100)
CT3	Option HT : Multiplication factor applied to V3 in % (0..100)
CT4	Option HT : Multiplication factor applied to I1 in % (0..100)
CT5	Option HT : Multiplication factor applied to I2 in % (0..100)
CT6	Option HT : Multiplication factor applied to I3 in % (0..100)

(Value) is an four digit integer left padded with zeroes.

(1) The RTI value corresponds to the value of the relation of the actual current transformer. That value is the same that must be entered in the Circutor, so that its reading correspond con real values of the currents in the lines.

(2) (3)(4) See description of Option HT

Possible answers:

All OK (ACK) “>(OldAddr)(Checksum)<CR>”

Error (NACK) “?(Addr)(Checksum)<CR>”

3. Command for reading the programmed values of the signals:

General format: “\$(Addr)R(Variable)(Checksum)<CR>”



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Where (Variable) is one of the Signals name indicated in the table above.

Possible answers:

If no errors: OK (ACK) ">(Value)(Checksum)<CR>"

where (Value) is the value read, four digits

If errors (NACK) "? (Addr)(Checksum)<CR>"

Special commands:

4. Calibration using the actual settings

"#(Addr)C(Checksum)<CR>"

Possible answers command C:

If no errors OK (ACK) ">(Status)(Checksum)<CR>"

If errors (NACK) "? (Addr)(Status)(Checksum)<CR>"

Where (Status) is as 8 ASCII characters "0", "1" or "2" with the following meaning:

First two digits indicate the communication status of the connection between the microcontroller with the measurement unit (internal instrument).

Possible values:

0 communication OK.

1 timeout waiting for the instrument.

2 Checksum error in the answer.

The first of of these two digits are for checking the voltage generators, the second for the current generators.

The remaining six digits indicate the result of the calibration process for each generator in the order V1, V2, V3, I1, I2, I3.

Possible values:

0 Calibration OK. Associated coefficient adjusted accordingly.

1 Out of range. Unable to adjust. Excessive difference with the expected value.

2 Variable level too low.

Example:

Cod: 00 000 111 Conversion out of range in all the current channels
(check the transformers conversion factor)



5. Save configuration in the internal E2PROM

“#(Addr)G(Checksum)<CR>”

6. Software Reset

“ # (Addr)R(Checksum)<CR>”

7. Query equipment identification

“#(Addr)I(Checksum)<CR>”

Answer command I:

“>GPTR <serial number> <PartNumber> /<Hardware Version> <Software Version>(Checksum)<CR>”

8. Query Hardware Status

“#(Addr)E(Checksum)<CR>”

Answer to command E:

“>EE <Volts at 5VDC rail> <Volts at 24VDC rail>(Checksum)<CR>”

where:

EE = Status byte in hexadecimal with the following meaning for the bits combinations

Bits	Applies to
1 and 0	Last reset code
2	E2Prom status
4 and 3	RS485 communication status
6 and 5	Calibration instrument communication status
7	Over current

Last reset code

'00' → OK

'01' → Reset SW

'10' → Brown Out Voltage raised (power supply too low)

'11' → Internal "Watch Dog Timer"

E2Prom



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'1' → Checksum error reading the stored configuration:

Node number, calibration coefficients, current transformers conversion factor,..

Host communication status.

'00' → OK

'01' → OverRun

Calibration instrument communication status.

'00' → OK

'01' → Timeout

'10' → Checksum error

'11' → Invalid data

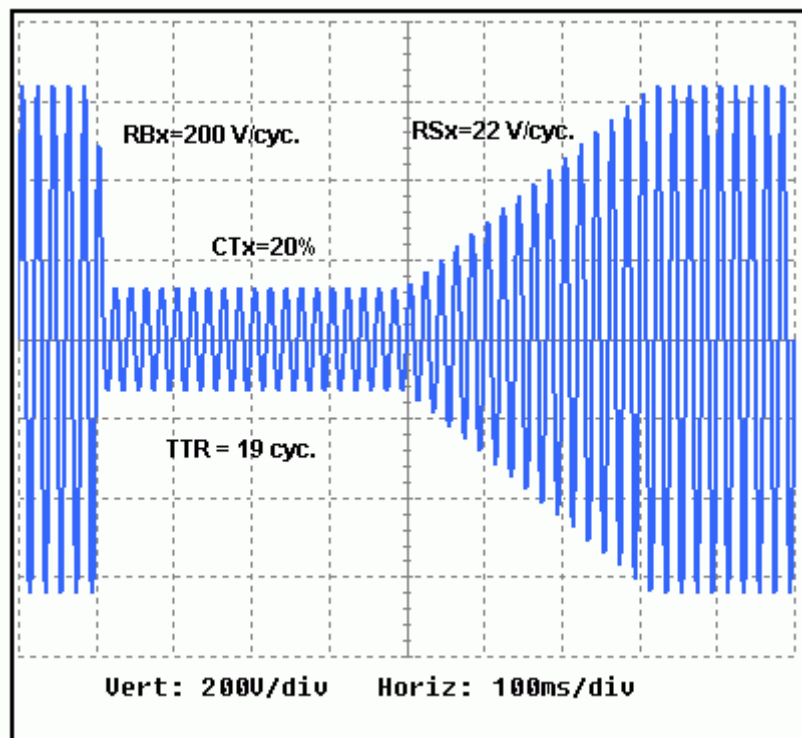
OverCurrent

'1' → At least one channel has been disconnected by the automatic overcurrent protection. It's necessary to issue a reset command to restart normal operation.



9. Option HT

This option is a facility implemented to facilitate the easy specification and execution of “holes in the grid”, according with the typical specification parameters.



In order to use this functionality the controlling program should program first the parameters (RBx, TTR, RSx) and then issue the command

“ #(Addr)T(Checksum)<CR>” Generates de current preprogramed transient



Some examples

In the next example the value of the RS485 address is 00. Values for the checksum are not shown

Example 1: Generation of Slow ramps

‘ First set the rate of rising value

\$00WRS10005 ‘ Rising ramp channel 1 (V1) at = 5 V/Cycle

‘ Then set the desired final value

\$00WVE10390 ‘ V1 will grow to 390V rms in around 2 seconds

‘Set the falling ramp

\$00WRB10001 ‘ Falling ramp V1 at 1 V/Cycle

‘ Set the new value

\$00WVE10000 ‘V1 will drop to 0 V rms At 1V/Cycle takes (around 7,8 seconds)

‘ Similar case for I1

\$00WRS40001 ‘Rising ramp channel 4 (I1) at = 1 A/Cycle

\$00WIE11000 ‘ Set I1=1000A takes 20 seconds

Example 2: Transient generation (Option HT):

Generation of a “hole” in the grid:

#00R Reset

\$00WRB11000 ‘ Falling ramp max in all channels

\$00WRB21000

\$00WRB31000

\$00WRB41000

\$00WRB51000

\$00WRB61000

\$00WRS10015 ‘ Rising ramp calculated for 0.5s up to 390V/1000A

\$00WRS20015

\$00WRS30015



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\$00WRS40040

\$00WRS50040

\$00WRS60040

\$00WVE10390 ‘ Set all the voltages to 390 V

\$00WVE20390

\$00WVE30390

\$00WTTR0025 ‘ Transient duration 25 cycles (0.5s)

#00T ‘ Generate the transient

Example 3: Transient generation (Option HT) with currents:

Similar above case but with currents

\$00WIE11000

\$00WIE11000

\$00WIE11000

#00T ‘Generate the transient

Specifications for the change of values

The GPTR includes a protection mechanism for hazard consisting in a trip sensor for each of the grid generators (V1,V2,V3). The level of this trip is adjusted to 12 mA.

That has some implications in the values that can be used in the rate of change of values. That is due to the fact that the actual current limit will depend on the kind of load connected to V1, V2, V3. On top of that, there is transformer in each of these signals that can greatly influence the maximum rate of change that actual can be applied before the trip mechanism enter in action.

Software drivers

Beside the low level commands described above, ACM provides a DLL that present an easy to use interface for standard programming languages (Visual C, VB, Delphi, etc).

This driver controls one of the COMx ports of a host PC, avoiding the need to worry about low level channel details.

Control Program

The GPTR comes with a control program that allows the setting of values in every generator.



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GPTR Controller (version for 5A current generators)

Exit User interface Miscellaneous

☒ All together

Grid Voltages

V1
V2
V3

Phase angles
Phase 2
Phase 3

Current generators (mA)

☒ All together ☒ All together Delta phase

I1
I2
I3

Phase 1 →
Phase 2 →
Phase 3 →

Frequency Hz

Grid Analyzer

V

Ramp Up (V/cycle)

RS1
RS2
RS3

Ramp Down (V/cycle)

RB1
RB2
RB3

Down Transient(%)

CT1
CT2
CT3

☒ All together ☒ All together ☒ All together

mA

RS4
RS5
RS6

RB4
RB5
RB6

CT4
CT5
CT6

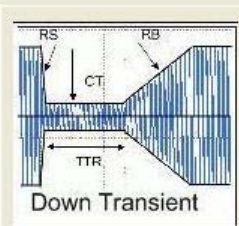
TTR

Sheet values = GPTR Internal values

Calibration Store parameters GPTR Reset

Generators Online Grid Analyzer Online

Communications
COM3

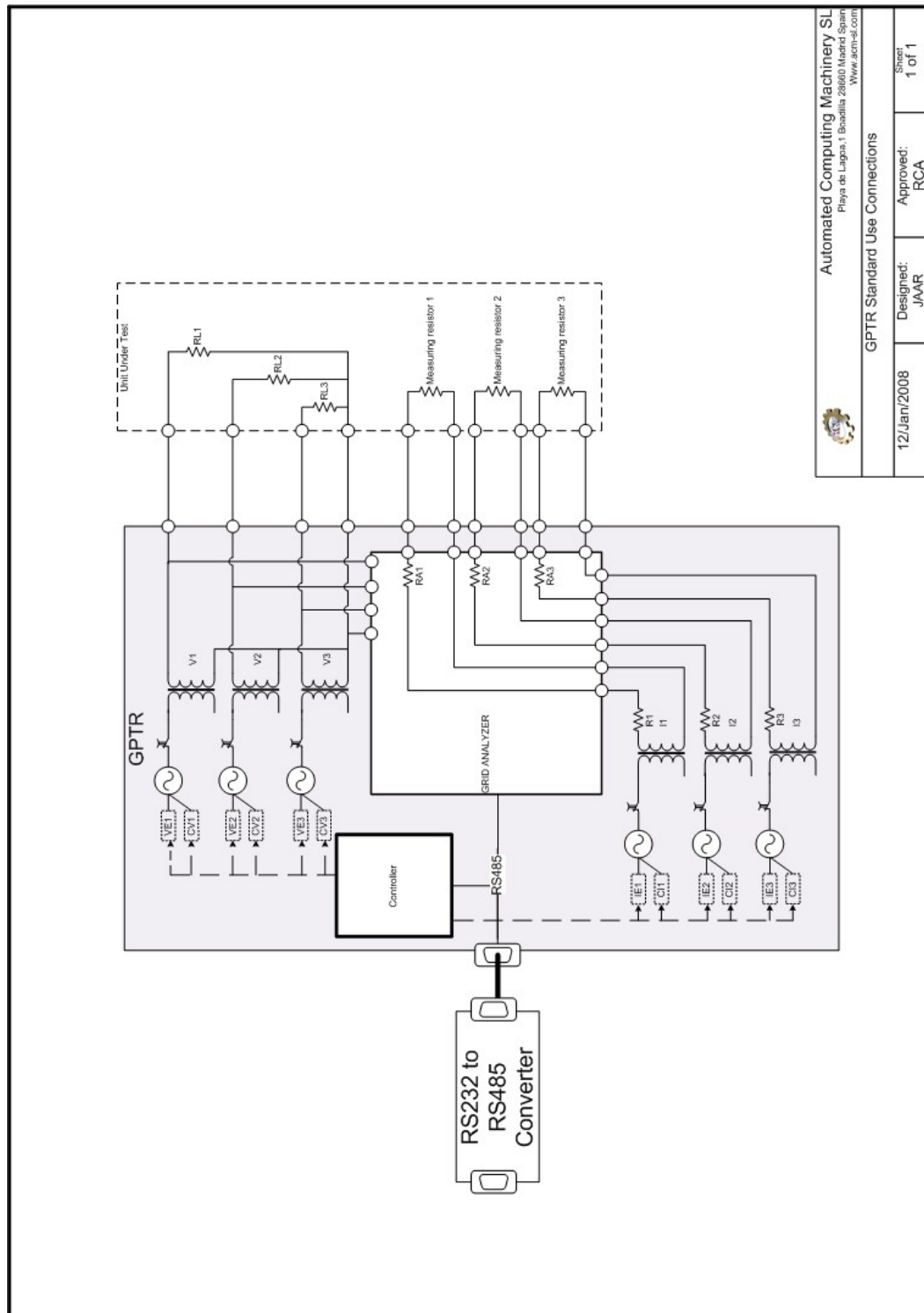




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Connection



 Automated Computing Machinery SL Playa de Lagoa, 1 Boadilla, 28660 Madrid Spain www.acm-sl.com			
GPTR Standard Use Connections			
12/Jan/2008	Designed: JAAR	Approved: RCA	Sheet 1 of 1



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Applications

Test of Grid Analyzers

The GPTR is well suited for reproducing the conditions for testing Grid Analyzers, both in the voltage inputs as well as in the current inputs and power factor for each voltage/current pair.

Those conditions can be set up in the 4 quadrants (both supply and consumption).

See the “Connection” to get a typical setup for this application.

The capability for self-calibration using the internal Grid Analyzer, and the excellent linearity of the internal control in the generators makes the GPTR an ideal tool for exploring all the situations a Grid Analyzer may encounter.

Test of Algorithms in PLCs

For PLCs, both stand-alone and embedded in service panels of big industrial installations, the GPTR is an ideal tool to check the dynamic behavior of the implemented algorithms, in conditions very close to reality, but in a controlled and secure environment, prior to commissioning the whole control.

The command set of the GPTR is designed so that it is possible to set for conditions and evolution of those conditions in a very easy way, but inside an intrinsically safe environment. The wide operative ranges of the generators inside the GPTR and the capability for controlling the changes of the variables, make the GPTR an ideal tool for reproducing hard-to-create conditions (frequency changes, power factor changes, unbalanced situations, etc)

Test of Service Panels

In Service Panels of medium or high complexity that incorporate Grid Analyzers, control units (i.e. PLCs), sophisticated Human Machine Interfaces, the GPTR can help as an ATE (Automatic Test Equipment) unit that ease the design of test strategies to cover the full spectrum required.